MD AL WALID

Working Student — Full-Stack Web Development

Vettersstraße 54, 09126 Chemnitz, Germany | +49 015566486259 | alwalidakash@gmail.com
linkedin.com/in/md-alwalid | Portfolio

PROFILE

Full-stack developer with a background in Information & Communication Engineering, currently pursuing a Master's degree in Web Engineering at TU Chemnitz. Nearly two years of professional experience as an Associate in an R&D team at a multinational company, developing strong problem-solving skills within a structured, collaborative engineering environment. Through hands-on projects, I have built full-stack web applications using modern technologies including React, Node.js, Express, and PostgreSQL/MongoDB. Seeking a working student (Werkstudent) position in web development to apply this experience alongside my studies, keep growing as a developer, and contribute to a team that values good engineering practices and continuous learning.

WORK EXPERIENCE

Associate — Research & Development Team

Quantanite (Multinational Company) | *Dec 2022 – Sep 2024* | *Dhaka, Bangladesh*

- Managed and maintained large-scale datasets to support AI training initiatives, ensuring high levels of data accuracy, consistency, and quality.
- Prepared and processed raw data according to project requirements while maintaining defined quality standards.
- Identified workflow challenges and implemented process improvements to increase efficiency, productivity, and overall output quality.
- Collaborated with internal teams and communicated directly with clients to understand requirements, resolve issues, and provide effective solutions.
- Trained and supported new team members by sharing process knowledge, quality guidelines, and best practices.
- Adapted workflows based on evolving business requirements while maintaining accuracy and timely delivery.

Intern Trainee — Network Operations Center

Bangladesh Submarine Cable Company Ltd. (BSCCL) | *Feb 2020 – Mar 2020* | *Kuakata, Bangladesh*

- Set up hardware and software configurations in the Network Operations Center to meet performance requirements.
- Helped formulate data center expenditure reports for management.

Industrial Trainee

Western Marine Shipyard Limited | *Aug 2016 – Nov 2016* | *Chattogram, Bangladesh*

- Gained practical experience in tool management and CNC machine operations.
- Monitored shipboard repair work to meet quality, safety, and efficiency standards.

FULL-STACK PROJECTS

Full-Stack E-Commerce Platform

Personal Project

https://ecommers-website-delta.vercel.app/

- Built a full-stack e-commerce platform with customer and admin roles, including registration, login, JWT authentication, and role-based authorization.
- Designed a normalized PostgreSQL schema with relationships, constraints, and indexes to support products, categories, carts, and orders.
- Developed a RESTful API with Node.js and Express, including product search, filtering, sorting, and pagination, request validation, and centralized error handling.
- Implemented shopping cart, checkout, order creation, and customer order history, plus an admin dashboard with platform statistics, product and category CRUD, and order status management.
- Containerized the application with Docker and set up GitHub Actions CI/CD pipelines for the frontend and backend, with automatic frontend deployment to Vercel.
- Tech stack: React, Bootstrap, Axios, Node.js, Express, PostgreSQL, JWT, bcrypt, Docker, GitHub Actions.

German Traffic Accident Analytics System (ETL Data Hub — Full-Stack Data Pipeline)

Personal Project

https://german-traffic-data.vercel.app/

- Built a full-stack ETL data pipeline extracting data from multiple public sources (CSV and ZIP archives) and loading it into a normalized PostgreSQL database.
- Designed and implemented data transformation logic (parsing, unpivoting, and cleaning) to convert raw datasets into structured, query-ready formats.
- Developed a RESTful API using Node.js and Express, exposing dedicated endpoints per dataset.
- Built a responsive React (Vite) dashboard with Bootstrap, featuring dataset selection, loading states, and error handling.
- Tech stack: Node.js, Express, PostgreSQL, React, Bootstrap, Axios, csv-parser, adm-zip.

AI-Powered Notes Management Application — Full-Stack Project

Personal Project

https://note-clener.vercel.app/

- Developed a full-stack notes management application with JWT-based authentication, protected routes, and CRUD operations.
- Integrated Google's Gemini API to provide AI-assisted grammar correction, text rewriting, professional tone adjustment, and note summarization.
- Designed a modular backend architecture using Express routes, controllers, and services to process AI requests.
- Implemented an AI suggestion workflow allowing users to review, accept, or discard generated text before saving notes.
- Containerized the React frontend and Express backend using Docker and Docker Compose for a consistent local development environment.
- Deployed the React frontend on Vercel and the Express backend on Render, using MongoDB for persistent storage.
- Tech stack: React.js, Node.js, Express.js, MongoDB, JWT, REST APIs, Gemini API, Docker, Docker Compose.

Zerodha Simple Clone — Full-Stack Trading Platform

Personal Project

https://zerodha-website-ecru.vercel.app/

- Built a full-stack, Zerodha-inspired stock trading platform using React, Node.js, Express.js, and MongoDB.
- Developed core trading features including holdings management, watchlist management, portfolio tracking, and a buy/sell order interface.
- Designed and integrated RESTful APIs (e.g. /allHoldings, /allPositions, /newOrder, /watchlist) to connect frontend and backend.
- Built a responsive landing page and interactive trading dashboard using React and Bootstrap, optimized for both desktop and mobile.
- Tech stack: React, Bootstrap, Axios, React Router, Node.js, Express.js, MongoDB.

SKILLS

Frontend: HTML, CSS, JavaScript, React, Bootstrap
Backend: Node.js, Express.js, RESTful API Design (CRUD Operations), Swagger/OpenAPI Documentation
Databases: PostgreSQL, MongoDB (used in projects), MySQL (basic)
Data Management: Data Accuracy & Quality Assurance, Large-Scale Dataset Management, AI Training Data Preparation
Programming Languages: C (basic), C++ (basic), Python (basic)
Soft Skills: Cross-Functional Collaboration, Client Communication, Process Improvement, Adaptability, Mentoring

EDUCATION

M.Sc. in Web Engineering

Technische Universität Chemnitz | *Oct 2024 – Present* | *Chemnitz, Germany*

B.Sc. in Information & Communication Engineering

Bangladesh Army University of Engineering & Technology (BAUET) | *2017 – 2022* | *Natore, Bangladesh*

Diploma in Marine Technology

City Engineering Institute | *2012 – 2017* | *Khulna, Bangladesh*

LANGUAGES

Bengali: Mother tongue
English: B2 (Independent user) — Listening, Reading, Spoken, Writing
German: A2 (Basic user)

ADDITIONAL

Organizing Secretary, BAUET ICT Club — coordinated club events and member communication (2021–2022)